
FINAL PROJECT REPORT

CareerLens

Pakistan

AI-Driven Career Guidance for the Pakistani Professional Landscape

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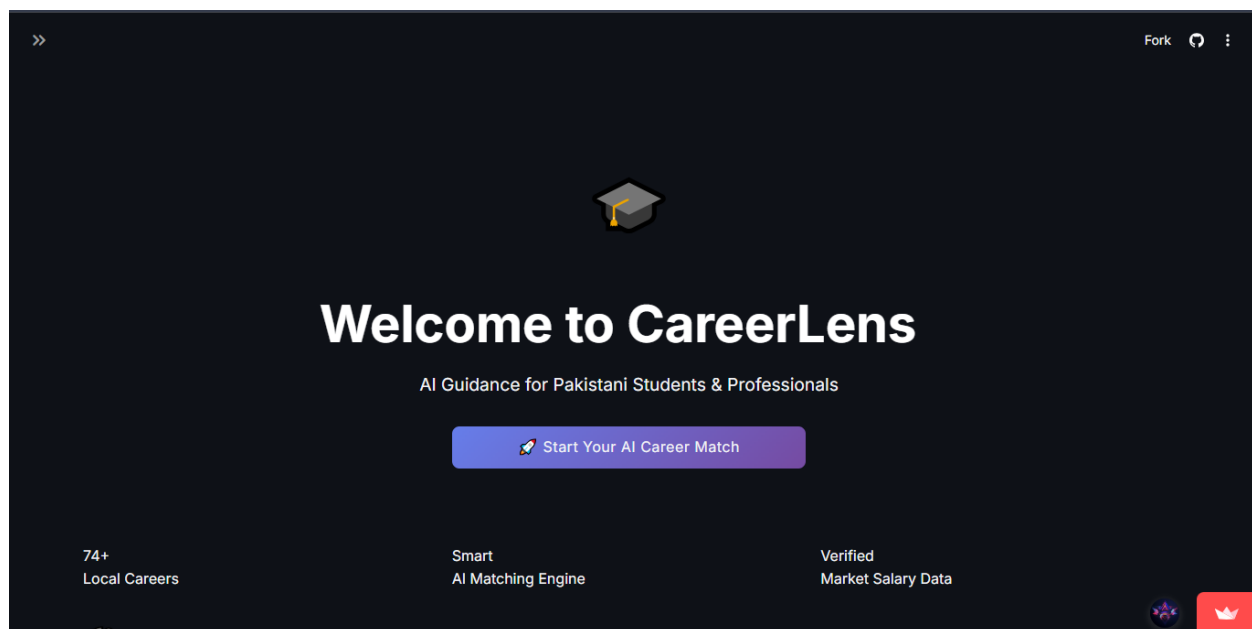
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1. Executive Summary

CareerLens Pakistan is an AI-powered, multi-page web application that bridges the gap between students' interests and Pakistan's evolving job market. By combining state-of-the-art Natural Language Processing (NLP) with supervised Machine Learning, the platform delivers personalized career matching, professional readiness assessments, and localized educational roadmaps — all through a clean, modern web interface built in Python and Streamlit.

The system draws on a curated database of 74+ locally relevant professions (ranging from MBBS and Civil Engineering to Blockchain Developer and DevOps Engineer) and a synthetic training dataset of 3,012 candidate profiles to produce data-driven career recommendations tailored to the Pakistani context.



2. Problem Statement

Students and young professionals in Pakistan face a significant information gap when making career decisions. Existing guidance tools are either too generic to be actionable or completely uninformed by local realities such as regional university availability, FSc/ICS intermediate pathways, and domestic salary benchmarks. The absence of structured, localized guidance frequently leads to mismatched career choices and underutilized talent.

CareerLens was designed specifically to address these gaps through three core capabilities:

- Localized AI matching against 74 Pakistani professions with regional university data.
- Predictive readiness evaluation calibrated to local professional entry standards.
- Structured educational roadmaps covering FSc/ICS tracks and provincial universities.

3. Project Objectives

The platform was built around four primary objectives:

Targeted Guidance	Provide career advice localized to Pakistani universities, FSc/ICS pathways, and 74+ domestic professions.
AI-Powered Matching	Use semantic text analysis and structured quizzes to match users to careers with human-readable confidence scores.
Readiness Evaluation	Predict a candidate's professional preparedness across five measurable dimensions using a regression model.
User Experience	Deliver a premium, minimalistic glassmorphism interface designed to reduce cognitive load.

4. System Architecture

The application follows a clean four-layer modular architecture, ensuring clear separation of concerns, independent testability, and easy extensibility. Each layer has a single, well-defined responsibility.

4.1 Frontend — `app.py` + `pages/`

The frontend layer is built entirely on Streamlit's modern `st.navigation` and `st.Page` API, which provides seamless client-side multi-page routing. The entry point (`app.py`) initializes the application in wide layout mode and orchestrates navigation between five pages: Home, AI Matcher, Readiness Assessment, Career Directory, and Career Comparison. The Home page displays platform metrics and provides one-click navigation buttons to each module.

4.2 AI Core — `ml_engine.py`

This is the computational brain of the application. All models and large datasets are loaded once into memory using Streamlit's `@st.cache_resource` decorator, preventing repeated disk reads on every user interaction. The module manages the SBERT embedding model, the KNN classifier, the Label Encoder, and the Linear Regression readiness model. It exposes two key functions consumed by the page modules: `get_salary()` for salary retrieval with fallback logic, and `calculate_readiness()` for clamped regression output.

4.3 Design System — `ui_components.py`

This module encapsulates the entire visual identity of the platform. It injects custom CSS implementing a glassmorphism dark theme (background gradient: `#0f0c29` to `#24243e`), styled gradient buttons, and translucent card panels. The centerpiece is the `render_card()` function — a fully reusable component shared across the Matcher, Directory, and Compare pages — which

renders career title, category, match score, salary breakdown, and an expandable roadmap section consistently across the application.

4.4 Data Layer — data/ + models/

The data layer contains two CSV datasets and five serialized model files. The datasets provide the factual knowledge base (careers and training profiles), while the model files provide the pre-trained inference artifacts. The salary lookup files use a two-level JSON structure: career-specific data is tried first; if missing, the system falls back to the category average.

5. Technology Stack

The project was built entirely in Python on the following technology stack:

Layer	Technology	Purpose
Web Framework	Streamlit 1.57+	Multi-page UI routing, layout, and widget system
NLP	Sentence-BERT (all-MiniLM-L6-v2)	Semantic text encoding into 384-dimensional vectors
ML — Classification	Scikit-learn KNN	Career category prediction from embedded query vectors
ML — Regression	Scikit-learn Linear Regression	Professional readiness score prediction (0–100%)
Visualization	Plotly, Altair	Interactive gauge charts and grouped salary bar charts
Data Processing	Pandas, NumPy	Dataset loading, missing value handling, computation
Persistence	Joblib (.pkl), JSON	Model serialization and salary lookup tables
Dev Environment	.devcontainer (Python 3.11)	Reproducible setup via VS Code / GitHub Codespaces

6. Project File Structure

The project follows a logical, modular directory layout:

app.py	Application entry point — initializes Streamlit, configures routing.
ml_engine.py	All ML logic — model loading, embedding computation, salary lookup, readiness prediction.
ui_components.py	CSS design system and the shared <code>render_card()</code> component.

<code>data/careers_pakistan.csv</code>	74 curated careers with roadmaps, salaries, degrees, and university data.
<code>data/careerlens_training_dataset.csv</code>	3,012 synthetic training profiles mapped to careers and categories.
<code>models/knn_model.pkl</code>	Serialized KNN classifier.
<code>models/readiness_model.pkl</code>	Serialized Linear Regression readiness model.
<code>models/career_embeddings.npy</code>	Pre-computed SBERT vector matrix for all 74 careers.
<code>models/salary_lookup.json</code>	Career-level salary data (Min/Avg/Max).
<code>models/category_salary_fallback.json</code>	Category-level salary averages for missing career entries.
<code>pages/ai_matcher.py</code>	Semantic search and Career Compass Quiz feature.
<code>pages/readiness.py</code>	Readiness assessment with gauge chart and actionable feedback.
<code>pages/directory.py</code>	Filterable career catalog.
<code>pages/compare.py</code>	Side-by-side career comparison with charts and tables.
<code>.devcontainer/</code>	Docker/Codespaces configuration for reproducible development.

7. Application Features

7.1 AI Matcher — Semantic Career Search

The flagship feature of CareerLens. It provides two distinct pathways for users to discover matching careers, both feeding into the same underlying ML pipeline.

Path A — Free-Text Semantic Search

The user types a natural-language description of their interests (e.g., 'I enjoy building websites and solving logic problems'). The system first sanitizes the input using regular expressions to block gibberish and keyboard-smash inputs. The cleaned text is then encoded by the SBERT

model into a 384-dimensional dense vector. Cosine similarity is computed between this query vector and all pre-computed career embedding vectors to rank career matches.

Path B — Career Compass Quiz

A four-question interactive quiz displayed using Streamlit's `st.pills` widget (clickable option chips). The four dimensions covered are: favourite academic subjects, preferred problem-solving style, desired work environment, and primary career goal. Each selection is mapped to a keyword string through a pre-defined expansion dictionary, which is then fed into the same matching pipeline as free-text input.

The screenshot shows a Streamlit application titled "Find Your Dream Career". It has a dark theme and a sidebar on the right with "Fork" and other icons. The main content area has a sub-header "Choose how you want to find your match!" with two tabs: "Type Your Interests" and "Take the Career Compass Quiz". The "Take the Career Compass Quiz" tab is active. Below it is a section titled "The Career Compass" with the instruction "Answer these 4 questions to let our AI build your perfect professional profile." The first question is "1. Which topics naturally fascinate you?" with five options: "Computers, Software & Tech", "Physics, Machines & Engineering", "Biology, Anatomy & Medicine", "Art, Design & Digital Media", and "Business, Markets & Economics". The second question is "2. How do you prefer to solve problems?" with five options: "Deep Focus / Solo Coding", "Analyzing Data / Researching", "Brainstorming / Creative Thinking", "Leading / Managing Teams", and "Directly Helping/Treating People". The third question is "3. What is your dream work environment?" with five options: "A modern tech office or remote from home", "A hospital, clinic, or medical lab", "A corporate high-rise or financial firm", "Outdoors, on a site, or travelling", and "A creative studio or production set". The fourth question is partially visible at the bottom.

Matching Pipeline (Both Paths)

Both input paths execute the same four-stage pipeline:

- The KNN classifier predicts the broad career category (e.g., 'IT & Data', 'Medical') from the encoded vector.
- Cosine similarity identifies the Top 3 specific career matches within the full database.
- Raw cosine scores (typically 0.22–0.60+) are scaled to human-readable percentages (50%–99%) using a non-linear mapping algorithm.
- An execution gate rejects inputs with a raw top score below 0.22 as 'Out of Scope', preventing irrelevant or misleading results from reaching the user.

7.2 Readiness Assessment

A predictive self-evaluation tool that estimates a user's professional readiness as a percentage score. The user rates themselves on five dimensions using sliders (scale 0–10):

- **Academic Performance:** GPA or general academic achievement.
- **Technical Skills:** Domain-specific technical knowledge.
- **Extracurricular Involvement:** Participation in projects, societies, or competitions.
- **Communication Skills:** Written and verbal professional communication.
- **Career Goal Clarity:** Awareness and direction regarding professional objectives.

Career Readiness

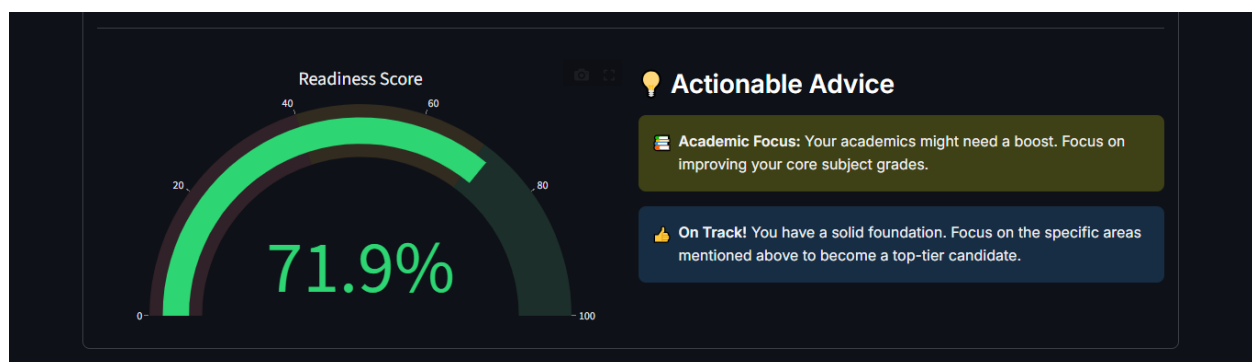
Rate your skills on 5 key dimensions to see your overall career readiness percentage.

Self-Assessment Form

Dimension	Score
Academic Performance	5
Technical Skills	8
Extracurricular Activities	7
Communication / Soft Skills	7
Career Goal Clarity	8

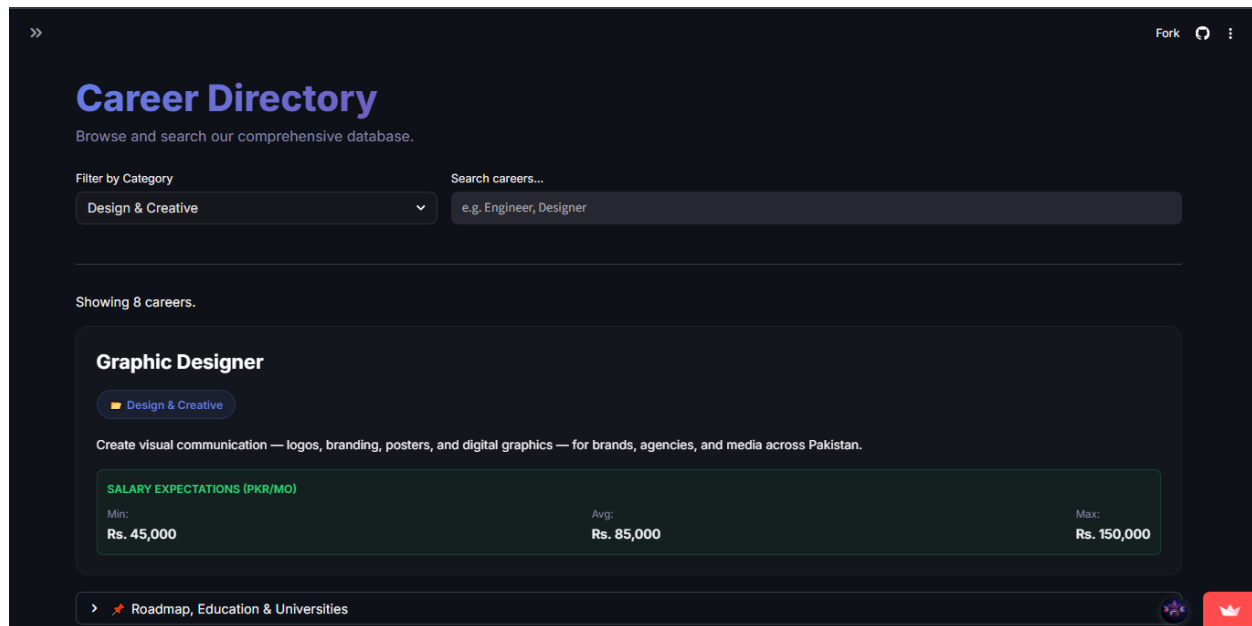
[Calculate Readiness Score](#)

These five values are passed to the pre-trained Linear Regression model, which outputs a predicted readiness score clamped to the 0–100% range. The result is rendered as an interactive Plotly Gauge Chart that dynamically changes color: red for low readiness (<40%), yellow for moderate (40–70%), and green for high (>70%). Below the chart, conditional logic generates specific, personalized feedback — for example, if Technical Skills is rated below 6, the system recommends relevant certifications or coding platforms.



7.3 Career Directory

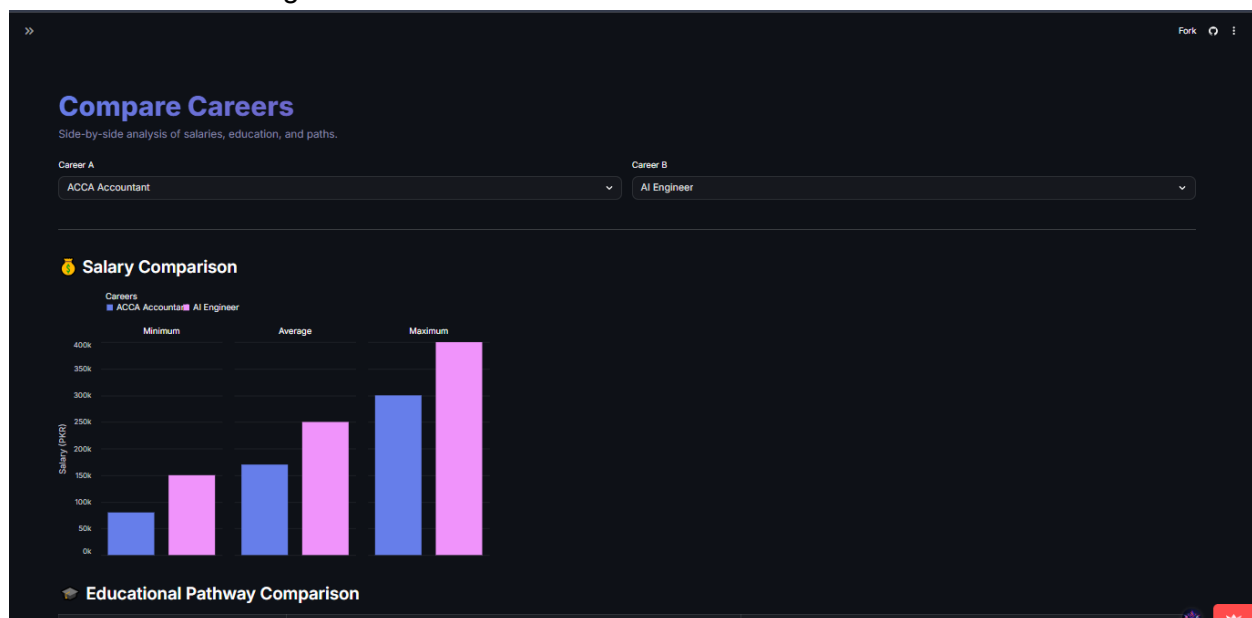
A fully searchable and filterable catalog of all 74 careers in the database. Users can filter by any of 8 predefined categories (IT & Data, Engineering, Medical & Health, Business & Finance, Law & Policy, Education, Creative Arts, Sciences) and search by free-text keyword. Each result is rendered using the shared `render_card()` component, ensuring visual consistency. Expandable card sections reveal the full step-by-step roadmap, required degree, intermediate pathway (FSc or ICS), and top universities distributed across 5 provinces.



7.4 Career Comparison

Enables side-by-side analysis of any two careers selected from dropdown menus. The comparison view provides two levels of insight:

- Salary Comparison — An Altair grouped bar chart visualizes Minimum, Average, and Maximum salary for both careers side by side, making relative earning potential immediately legible.
- Education Comparison — A clean HTML table contrasts the required degrees, recommended intermediate pathways (FSc/ICS), and core skill sets for both professions in a structured grid.



8. Datasets and Model Details

8.1 Careers Dataset — careers_pakistan.csv

The primary knowledge base of the system. Contains 74 rows of detailed career data covering both traditional professions (MBBS, Civil Engineering, Law) and modern technology roles (Blockchain Developer, DevOps Engineer, Data Scientist). Each row contains: career name, category, minimum/average/maximum salary ranges (PKR), a multi-step roadmap, required degree, recommended FSc/ICS pathway, core skills list, and top universities distributed across Punjab, Sindh, KPK, Balochistan, and the federal capital. Missing values are handled at load time using Pandas fillna() to prevent runtime errors.

8.2 Training Dataset — careerlens_training_dataset.csv

Contains 3,012 synthetically generated candidate profiles used to train the KNN classification model. Each row represents a fictional candidate described by their skills, interests, and hobbies — mapped to a recommended career name and career category. The synthetic generation ensures balanced coverage across all 8 categories and avoids data sparsity for less common professions.

8.3 Trained Models

Model	Type	Role
SBERT (all-MiniLM-L6-v2)	Transformer (NLP)	Encodes input text and career keywords into 384-dimensional semantic vectors
KNN Classifier	Scikit-learn	Predicts the broad career category from an encoded query vector
Label Encoder	Scikit-learn	Encodes and decodes career category string labels for the KNN pipeline
Linear Regression	Scikit-learn	Predicts readiness score (0–100%) from 5 self-rating inputs
Career Embeddings	NumPy Array (.npy)	Pre-computed 384-dim SBERT vectors for all 74 careers — enables fast cosine matching

9. HCI and UX Design Decisions

The interface was designed following Human-Computer Interaction (HCI) principles that prioritize clarity, efficiency, and reduced cognitive load — particularly for users who may not be technically proficient.

Cognitive Load Reduction	Raw cosine similarity scores are never shown to users. Only scaled, human-readable match percentages (50%–99%) are displayed.
Modern Interaction	Traditional dropdown forms replaced with <code>st.pills</code> (clickable chip buttons) and sliders, reducing friction in data entry.
Visual Communication	Gauge charts use a red-yellow-green color system to communicate readiness at a glance, without requiring users to interpret raw numbers.
Design Consistency	The <code>render_card()</code> component ensures every career entry — regardless of which page it appears on — is displayed in the same structured, scannable format.
Glassmorphism Theme	A dark gradient base (<code>#0f0c29</code> to <code>#24243e</code>) with translucent card panels provides a polished, modern aesthetic while maintaining text readability.

10. Setup and Deployment

10.1 Development Environment

The project ships with a `.devcontainer.json` configuration that allows any developer to open the project in GitHub Codespaces or Visual Studio Code and have all dependencies — including Python 3.11 and all required packages — automatically installed with ports forwarded. This eliminates environment configuration errors and ensures consistent behavior across different machines.

10.2 Version Control

Git is used for version tracking throughout the project. A comprehensive `.gitignore` file prevents virtual environments (`venv/`), Python bytecode caches (`__pycache__`), and temporary build files from being tracked, keeping the repository clean and portable.

10.3 Running the Application

Once the environment is active, the application is launched with a single command:

```
| streamlit run app.py
```

Streamlit automatically opens the application in the default browser on localhost.

11. Conclusion and Future Scope

11.1 Conclusion

CareerLens Pakistan successfully demonstrates how modern AI techniques can be localized to solve a specific, high-impact problem in Pakistan's education and career counseling space. The platform integrates semantic text embeddings (SBERT), KNN classification, and linear regression into a cohesive, user-friendly interface — making sophisticated AI recommendations accessible to students with no technical background.

Key engineering strengths include the execution gate for out-of-scope filtering, the non-linear confidence scaling for human-readable scores, the two-level salary fallback mechanism, and the @st.cache_resource pattern for efficient model serving. Together, these produce a system that is both robust and fast in practice.

11.2 Future Scope

The following enhancements are planned for future development iterations:

- Real-time job postings — API integration with local portals such as Rozee.pk and LinkedIn Pakistan to show live vacancies for matched careers.
- Expanded training dataset — Growing the synthetic training data beyond 3,012 records to improve KNN category prediction accuracy for edge-case profiles.
- Mobile-native application — A dedicated iOS/Android app for broader accessibility, particularly for students in regions with limited desktop access.
- Counselor dashboard — An administrative view for career counselors to review aggregated anonymized interest data across institutions.
- University application guidance — Integration of admission deadlines, entry test requirements (MDCAT, ECAT, NTS), and merit thresholds for top universities.

11.3 GitHub Repo:

<https://github.com/abuba-akar0/CareerLens.git>

11.4 WebApp Link:

<https://careerlens-pakistan.streamlit.app/>